

6.1 Deformation: Stress & Strain

Question Paper

Course	CIEA Level Physics
Section	6. Deformation of Solids
Topic	6.1 Deformation: Stress & Strain
Difficulty	Medium

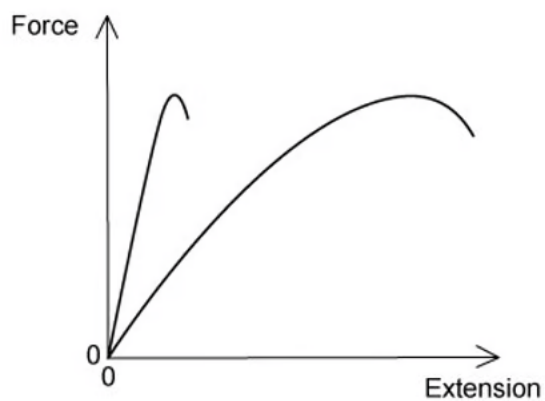
Time allowed: 20

Score: /10

Percentage: /100

Question 1

Two materials with the same dimensions were loaded to fracture; the graph shows the force-extension of these.



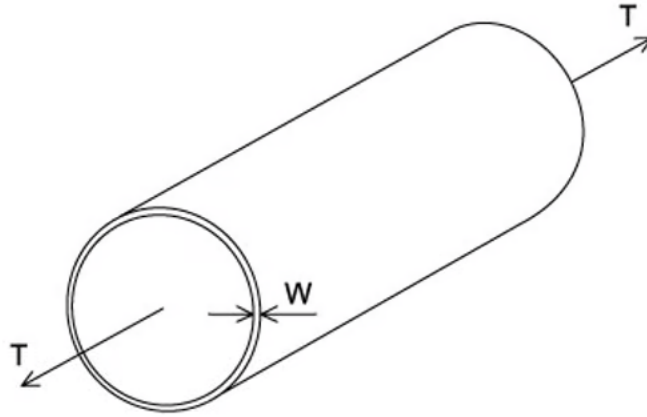
Which of the following would describe the behaviour of the materials?

- A. both materials are plastic
- B. both materials have the same ultimate tensile stress
- C. both materials are brittle
- D. both materials obey Hooke's law

[1 mark]

Question 2

A metal tube with a thin wall thickness w is shown in the diagram. The thickness of the wall is small when compared to the diameter of the tube.



A force T applied parallel to the axis of the tube puts the tube under tension. It is proposed that making the wall thicker would reduce the stress on the tube.

If the tube diameter and the tension remain the same, which wall thickness would half the stress?

- A. $4w$
- B. $2w$
- C. $\sqrt{2w}$
- D. $\frac{w}{2}$

[1 mark]

Question 3

A steel metal wire has the following properties:

$$\text{diameter} = 5.0 \times 10^{-4} \text{ m}$$

$$\text{Young modulus} = 2.0 \times 10^{11} \text{ Pa}$$

$$\text{tension} = 20 \text{ N}$$

The string snaps and contracts elastically.

By what percentage does a length l of a piece of the string contract?

- A. $1.3 \times 10^{-2}\%$
- B. $1.3 \times 10^{-4}\%$
- C. $5.1 \times 10^{-4}\%$
- D. $5.1 \times 10^{-2}\%$

[1 mark]

Question 4

The Young modulus of a metal wire is dependent on which property?

- A. spring constant
- B. ductility
- C. ultimate tensile stress
- D. elastic limit

[1 mark]

Question 5

The Young modulus E and the force per unit extension k describes the behaviour of a wire under tensile stress.

For a wire of length L and cross-sectional area A , what is the relationship between E and k ?

A. $E = \frac{A}{kL}$

B. $E = \frac{kA}{L}$

C. $E = \frac{kL}{A}$

D. $E = \frac{L}{kA}$

[1 mark]

Question 6

A steel bar, shown in the diagram, has a circular cross-section that is under tension T .



The diameter of the thinner portion is half the diameter of the wider portion.

What is the value of $\frac{\text{stress on the wide portion}}{\text{stress on the narrow portion}}$?

A. 4.0

B. 2.0

C. 0.50

D. 0.25

[1 mark]

Question 7

A student wanted to find the young modulus of a wire. The equation for the young modulus E is:

$$E = \frac{4Fl}{\pi d^2 x}$$

The student extended the wire with a known force and made a series of measurements.

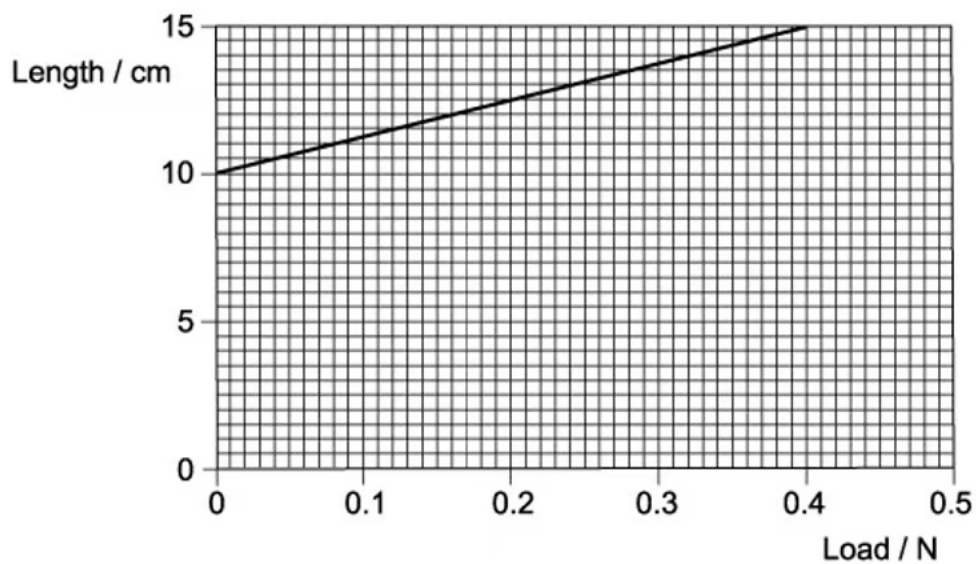
Which measurement has the largest effect on the uncertainty in the value of the calculated Young modulus?

	Measurement	Symbol	Value
A	length of wire before force applied	l	$2.043 \pm 0.002 \text{ m}$
B	force applied	F	$19.62 \pm 0.01 \text{ N}$
C	extension of wire with force applied	x	$5.2 \pm 0.2 \text{ mm}$
D	diameter of wire	d	$0.54 \pm 0.02 \text{ mm}$

[1 mark]

Question 8

A spring was stretched with increasing load. The graph of the results is shown below.



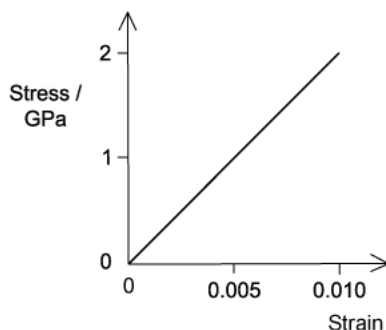
What is the spring constant?

- A. 0.080 N m^{-1}
- B. 0.13 N m^{-1}
- C. 2.7 N m^{-1}
- D. 8.0 N m^{-1}

[1 mark]

Question 9

The graph below shows the stress-strain for a metal.



When the strain on the rod is 0.010, what is the strain energy per unit volume of a rod made from this metal?

- A. 10 kJ m^{-3}
- B. 10 MJ m^{-3}
- C. 100 kJ m^{-3}
- D. 1.0 MJ m^{-3}

[1 mark]

Question 10

A steel spring was stretched to a length of 55 cm when a 25 N weight was hung from it. The spring has a spring constant of 150 N m^{-1} .

What was the original length of the spring?

- A. 0.61 m
- B. 0.72 m
- C. 0.38 m
- D. 0.49 m

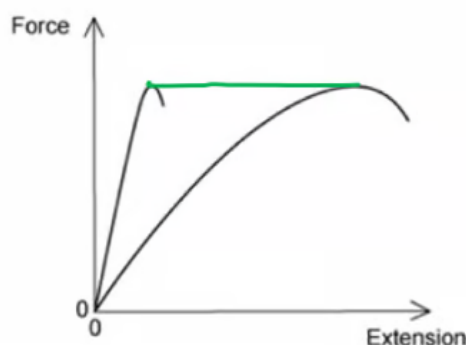
[1 mark]

Model Answers: Medium

1

The correct answer is **B** because:

- The ultimate tensile strength is the maximum stress a material can withstand before breaking
 - $\text{Stress} = \frac{\text{force}}{\text{area}} = \frac{F}{A}$
- From the graph, the breaking point of both materials is about the same force (draw a horizontal line from the maxima of both curves)



- The question states that both materials have the same dimensions, so both have the same ultimate tensile stress

A is incorrect because a plastic material undergoes permanent deformation, but doesn't break; the second line shows plastic behaviour

C is incorrect because a brittle material is one that doesn't show any deformation; the first line shows a brittle material

D is incorrect because a straight line shows Hooke's law, showing proportionality, these graphs curve and do not always obey Hooke's law

2

The correct answer is **B** because:

- The metal tube is hollow with the width of the wall w
- It is the area of the tube, not the hollow space that is taken into account
 - $\text{Stress} = \frac{\text{force}}{\text{area}} = \frac{F}{A}$
- Area of the wall = circumference $\times$ width = πdw
- As π and d are constant, the area of the wall is proportional to the width, w
- The tension force is kept constant, so the stress is inversely proportional to the area
 - $\text{Stress} \propto \frac{1}{\pi dw}$
- So, to half the stress, the thickness of the wall w would need to be doubled
 - Therefore, the new wall thickness would be $2w$

3

The correct answer is **D** because:

- The strain of the metal wire is equal to $\epsilon = \frac{\text{extension}}{\text{original length}} = \frac{\Delta L}{L}$
- The percentage of the length that contracts = strain $\times 100\%$
 - So, the percentage of the length that contracts = $\frac{\Delta l}{l} \times 100\%$
- The Young modulus is given by $E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\epsilon}$
 - Stress is equal to $\sigma = \frac{\text{force}}{\text{area}} = \frac{F}{A}$
- The cross-sectional area of the wire $A = \frac{\pi d^2}{4}$
- So, strain is equal to
 - $\epsilon = \frac{\sigma}{E} = \frac{1}{E} \left(\frac{F}{A} \right) = \frac{F}{E} \left(\frac{4}{\pi d^2} \right)$
 - $\epsilon = \frac{20}{2.0 \times 10^{11}} \times \frac{4}{\pi (5.0 \times 10^{-4})^2} = 5.09 \times 10^{-4}$
- So, as a percentage this is:
 - Strain $\times 100\% = 5.09 \times 10^{-4} \times 100\% = 5.09 \times 10^{-2} \%$
- Therefore, the percentage of the length that contracts is equal to $5.1 \times 10^{-2} \%$

4

The correct answer is **A** because:

- The Young modulus is given by $E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\epsilon}$
 - Stress is equal to $\sigma = \frac{\text{force}}{\text{area}} = \frac{F}{A}$
 - Strain is equal to $\epsilon = \frac{\text{extension}}{\text{original length}} = \frac{\Delta L}{L}$
- Therefore, Young modulus $E = \frac{(F/A)}{(x/L)} = \frac{FL}{Ax}$
- Hooke's law states $F = ke$, where k is the spring constant
- A metal wire will obey Hooke's law up to a point, the spring constant depends on the force, and this can be found in the equation for the Young modulus:
 - $E = \frac{(ke)L}{Ax} = \frac{kL}{A}$
- So, Young modulus E depends on the **spring constant** (and original length L and cross-sectional area A)

B is incorrect because not all materials are ductile, but all materials can have a Young modulus

C is incorrect because the ultimate tensile strength is the maximum stress a material can withstand before breaking; the Young modulus does not depend on this

D is incorrect because the elastic limit is beyond the limit of proportionality; the Young modulus does not depend on this

5

The correct answer is **C** because:

- The Young modulus is given by $E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\epsilon}$
 - Stress is equal to $\sigma = \frac{\text{force}}{\text{area}} = \frac{F}{A}$
 - Strain is equal to $\epsilon = \frac{\text{extension}}{\text{original length}} = \frac{\Delta L}{L}$
- Therefore, Young modulus $E = \frac{(F/A)}{(x/L)} = \frac{FL}{Ax}$

- Hooke's law states that $F = kx$, so $k = \frac{F}{x}$
- Therefore, Young modulus is equal to
 - $E = \frac{FL}{Ae} = \frac{L}{A} \left(\frac{F}{e} \right) = \frac{kL}{A}$

6

The correct answer is **D** because:

- Stress is equal to $= \frac{\text{force}}{\text{area}} = \frac{F}{A}$
- The steel tube is thicker at one end by double that of the thinner end
- The tension force is kept constant, so the stress is inversely proportional to the area
- The ratio of the stress in the two sections would be 0.25

Example of working:

narrow portion	wide portion
Diameter = 0.2 m, force = 10 N $\text{Area} = \pi r^2 = \pi \times 0.1^2 = 0.0314 \text{ m}^2$ $\text{Stress} = \frac{\text{force}}{\text{area}} = \frac{F}{A}$ $\text{Stress} = \frac{10}{0.0314} = 318.5 \text{ N m}^{-2}$	Diameter 0.4 m, force 10 N $\text{Area} = \pi r^2 = \pi \times 0.2^2 = 0.1256 \text{ m}^2$ $\text{Stress} = \frac{\text{force}}{\text{area}} = \frac{F}{A}$ $\text{Stress} = \frac{10}{0.1256} = 79.6 \text{ N m}^{-2}$

- So, the value of $\frac{\text{stress on the wide portion}}{\text{stress on the narrow portion}} = \frac{79.6}{318.5} = 0.25$

- The steel tube is thicker at one end by double that of the thinner end
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- So, the value of $\frac{\text{stress on the wide portion}}{\text{stress on the narrow portion}} = \frac{79.6}{318.5} = 0.25$

7

The correct answer is **D** because:

- The Young modulus is given by $E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\epsilon}$
 - Stress is equal to $\sigma = \frac{\text{force}}{\text{area}} = \frac{F}{A}$
 - Strain is equal to $\epsilon = \frac{\text{extension}}{\text{original length}} = \frac{\Delta l}{l}$
- The cross-sectional area of the wire $A = \frac{\pi d^2}{4}$
- So, the Young Modulus for a wire is given by
 - $E = \frac{4Fl}{\pi d^2 x}$

- The percentage uncertainty in the diameter needs to be doubled as it is squared in the equation

$$\circ \frac{\Delta E}{E} = \left(\frac{\Delta F}{F} \right) + \left(\frac{\Delta l}{l} \right) + 2 \left(\frac{\Delta d}{d} \right) + \left(\frac{\Delta x}{x} \right)$$

$$\blacksquare \left(\frac{\Delta F}{F} \right) = \frac{0.01}{19.62} = 0.00051$$

$$\blacksquare \left(\frac{\Delta l}{l} \right) = \frac{0.002}{2.043} = 0.00098$$

$$\blacksquare 2 \left(\frac{\Delta d}{d} \right) = 2 \left(\frac{0.02}{0.54} \right) = 0.07407$$

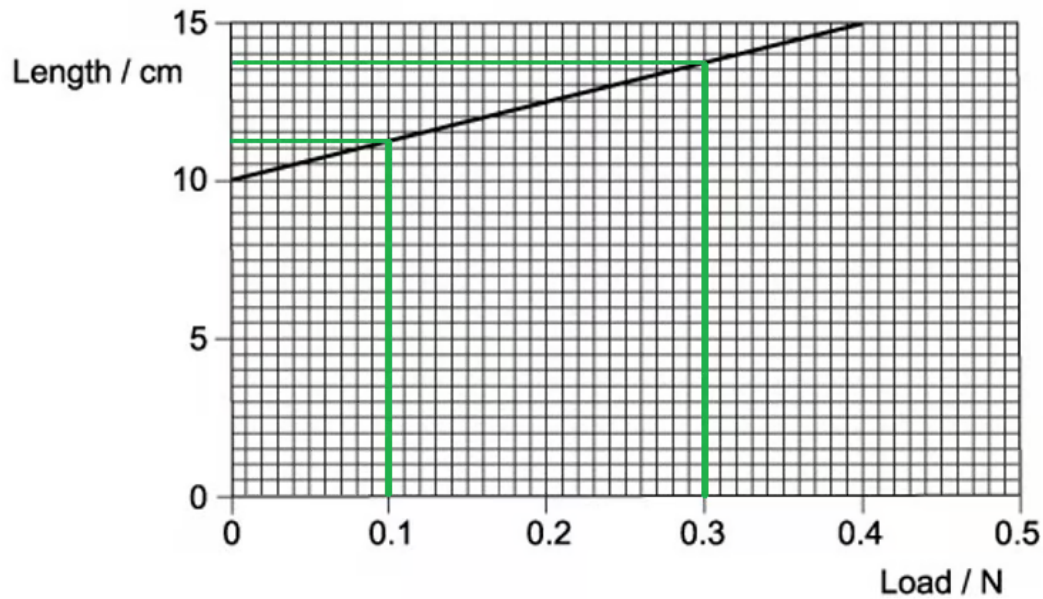
$$\blacksquare \left(\frac{\Delta x}{x} \right) = \frac{0.2}{5.2} = 0.03846$$

- Therefore, the diameter has the largest effect on the uncertainty of the Young modulus

8

The correct answer is **D** because:

- Hooke's law states $F = kx$, where k is the spring constant
- The gradient of a force-extension graph is the spring constant
 - Load = $0.3 - 0.1 \text{ N} = 0.2 \text{ N}$
- Extension of the spring $x = 13.75 - 11.25 = 2.5 \text{ cm} = 0.025 \text{ m}$
 - $k = \frac{F}{x}$
 - $k = \frac{0.2}{0.025} = 8.0 \text{ N m}^{-1}$
- Therefore, the spring constant is equal to 8.0 N m^{-1}



9

The correct answer is **B** because:

- When a material behaves elastically, the work done is stored as energy per unit volume – **strain energy**
- The strain energy is derived from the area under the stress-strain graph
 - Area under the graph = $\frac{1}{2} \times \text{base} \times \text{height}$
 - Strain energy = $\frac{1}{2} \times 0.010 \times 2 \times 10^9 = 1 \times 10^7 \text{ J m}^{-3} = 10 \text{ MJ m}^{-3}$
- Therefore, the strain energy is equal to 10 MJ m^{-3}

10

The correct answer is **C** because:

- Hooke's law states that $F = kx$, where k is the spring constant

Load / N

9

The correct answer is **B** because:

- When a material behaves elastically, the work done is stored as energy per unit volume – **strain energy**
- The strain energy is derived from the area under the stress-strain graph
 - Area under the graph = $\frac{1}{2} \times \text{base} \times \text{height}$
 - Strain energy = $\frac{1}{2} \times 0.010 \times 2 \times 10^9 = 1 \times 10^7 \text{ J m}^{-3} = 10 \text{ MJ m}^{-3}$
- Therefore, the strain energy is equal to 10 MJ m^{-3}

10

The correct answer is **C** because:

- Hooke's law states that $F = kx$, where k is the spring constant
- Let the original length = l
- When the force $F = 25 \text{ N}$, the extension $x = 55 \text{ cm} - l = 0.55 \text{ m} - l$
 - $F = k \times (0.55 - l)$
 - $\frac{F}{k} = 0.55 - l$
 - $l = 0.55 - \left(\frac{F}{k}\right) = 0.55 - \left(\frac{25}{150}\right) = 0.383 \text{ m}$
- Therefore, the original length of the spring is 0.38 m

